



Raabta

MULTI-TENANT WHATSAPP CRM

The shared WhatsApp inbox & CRM built for Pakistani SMEs.

Connect a business number by scanning a QR, and turn day-to-day WhatsApp chats into an organised, team-friendly sales & support workflow — in real time.

Domain

Conversational CRM / SaaS

Architecture

.NET 10 · Next.js 16 · Node

Status

MVP complete (Phases 0–7)

THE IDEA

One number, a whole team, zero chaos.

In Pakistan, WhatsApp is the storefront. SMEs run sales and support from a single phone — orders, payment screenshots, follow-ups, all buried in one chat list that only one person can see. Raabta turns that single number into a shared, multi-agent inbox with the structure of a real CRM, without asking customers to change anything about how they already message.

“Raabta” (رابطہ) means connection. The product keeps businesses and their customers connected — and keeps the team in sync behind the scenes.

8

PHASES SHIPPED

3

SERVICE TIERS

13+

DOMAIN MODELS

Realtime

LIVE INBOX

WHAT IT DOES

01 Shared team inbox

A 3-pane inbox where the whole team sees the same conversations update live over WebSockets — no more forwarding screenshots.

02 Assignment & workflow

Assign chats to agents, set statuses (Open / Pending / Closed), add private notes and colour labels, with role-based visibility.

03 Rich media

Send and receive images, video, voice notes and documents — with inline previews, a lightbox, and audio players.

04 Follow-ups & CRM

Contacts, reminders with due-date notifications, and canned quick replies — so no lead ever goes cold.

HOW IT WAS BUILT

Shipped **phase by phase.**

Each phase was designed, built, and verified end-to-end before the next — from the multi-tenant core to a production-shaped WhatsApp integration.

0

Foundation

Clean Architecture solution (Domain / Application / Infrastructure / API), PostgreSQL, and a typed Next.js front end.

1

Auth & multi-tenancy

JWT auth over httpOnly cookies, role-based access (Owner / Manager / Agent), invitations, and airtight per-tenant data isolation.

2

WhatsApp integration

Connect by QR; encrypted sessions stored in the database with auto-restore; HMAC-signed webhooks; live status over SignalR.

3

Realtime inbox

The shared 3-pane inbox: conversation list, message thread with optimistic send, and a contact panel — patched live as events arrive.

4

Team workflow

Assignment, statuses, internal notes, labels, agent-visibility rules, and an audit trail of who did what.

5

Reminders & quick replies

Follow-up reminders with a background due-sweep and in-app notifications, plus canned replies in the composer.

6

Dashboard & analytics

Workspace stats and team performance — with cards that deep-link straight into the underlying filtered views.

7

Rich media messaging

Full media pipeline — images, video, audio, voice notes, documents — with pluggable file storage behind a clean seam.

UNDER THE HOOD

Architecture & engineering.

BACKEND & DATA

ASP.NET Core · .NET 10

Clean Architecture

EF Core 10

PostgreSQL

SignalR

JWT

MailKit SMTP

WHATSAPP SERVICE

Node.js

Baileys

Express

AES-256 sessions

HMAC webhooks

FRONTEND

Next.js 16

React 19

Tailwind v4

Base UI

TanStack Query

PLATFORM

Multi-tenant

Realtime / WebSockets

HTTPS

Audit logging

ENGINEERING HIGHLIGHTS

- **True multi-tenancy.** A global query filter scopes every read and write to the current tenant automatically — isolation by default, not by discipline.
- **Resilient WhatsApp sessions.** Auth state is encrypted at rest and auto-restored on restart, so a redeploy never means re-scanning a QR.
- **Swappable transport.** An *IWhatsAppProvider* seam means the Baileys engine can be replaced with the official Meta Cloud API without touching the CRM.
- **Real-world correctness.** Handles WhatsApp's new LID addressing so replies actually deliver — the kind of detail that only surfaces with real users.
- **Live everywhere.** New messages, status changes and reminders are pushed to every connected teammate in under a second.

Built for the messy reality of how Pakistani businesses actually sell — on WhatsApp, as a team, all day.